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Exp_10

Title: Hamiltonian Cycles Directed Graph

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: n = int(input("Enter number of vertices in the graph: "))
print("Enter the adjacency matrix:")
graph = [list(map(int, input().split())) for _ in range(n)]

def find_hamiltonian_cycles(graph):
    n = len(graph)
    path = [0]
    visited = [1] + [0] * (n - 1)
    cycles = []

    def backtrack(v):
        if len(path) == n and graph[v][0]:
            cycles.append(path + [0])
            return

        for u in range(n):
            if graph[v][u] and not visited[u]:
                visited[u] = 1
                path.append(u)

                backtrack(u)

                path.pop()
                visited[u] = 0

    backtrack(0)
    return cycles

print("=====")
cycles = find_hamiltonian_cycles(graph)

if cycles:
    print("Hamiltonian Cycles found:")
    for cycle in cycles:
        print(" -> ".join(map(str, cycle)))
else:
    print("No Hamiltonian Cycle found")

print("=====")
```

Enter number of vertices in the graph: 5

Enter the adjacency matrix:

0 1 0 1 0

1 0 1 1 1

0 1 0 0 1

1 1 0 0 1

0 1 1 1 0

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Hamiltonian Cycles found:

0 -> 1 -> 2 -> 4 -> 3 -> 0

0 -> 3 -> 4 -> 2 -> 1 -> 0

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